Notes - LC passive wireless sensors using nonlinear parity-time symmetric configurations [1]

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September 1 2026 – October 8 2026

1 Preliminary Concepts [2]

1.1 Exceptional Points

Let us begin by understanding what exactly exceptional points are, and how they are useful in sensing.
First, let us look at an isolated system with two ideal oscillators, operating at frequencies ω_1 and ω_2 respectively. They are coupled symmetrically with a real coupling constant μ. If we say the complex amplitudes of each of the modes are a_1 and a_2 respectively, we can write the equations of motion as:

$$\begin{aligned} \dot{a}_1 &= -i\omega_1 a_1 - i\mu a_2 \\ \dot{a}_2 &= -i\omega_1 a_2 - i\mu a_1 \end{aligned} \tag{1}$$

In matrix form, we can write it as

$$\frac{d}{dt}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = -i\begin{pmatrix} \omega_1 & \mu \\ \mu & \omega_2 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \tag{2}$$

We can easily compare this equation to the Time-Dependent Schrödinger Equation (TDSE) $i\hbar\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle$. So, we can say our Hamiltonian is

$$\hat{H} = \begin{pmatrix} \omega_1 & \mu \\ \mu & \omega_2 \end{pmatrix} \tag{3}$$

We can clearly see that it is Hermitian.
Now, let us open up our system to the environment. Say their energy amplitudes decay at rates γ_1 and γ_2 respectively. If the oscillator is gaining energy from the environment (through a pump of photons in the photonic case, or a negative resistor in electronics), γ will be negative.
Due to this, the new complex amplitudes are

$$a(t) = a_0 e^{-i\omega t}e^{-\gamma t} = a_0 e^{-i(\omega - i\gamma)t} \tag{4}$$

Thus, the new Hamiltonian is

$$\hat{H} = \begin{pmatrix} \omega_1 - i\gamma_1 & \mu \\ \mu & \omega_2 - i\gamma_2 \end{pmatrix} \tag{5}$$

We see now that the Hamiltonian is *non-Hermitian.*
Our next step is to calculate the eigenvalues of this Hamiltonian.

$$\begin{aligned} \det(H - \sigma I) &= 0 \\ (\omega_1 - i\gamma_1 - \sigma)(\omega_2 - i\gamma_2 - \sigma) - \mu^2 &= 0 \\ \sigma^2 - \sigma[(\omega_1 - i\gamma_1) + (\omega_1 - i\gamma_2)] + [(\omega_1 - i\gamma_1)(\omega_2 - i\gamma_2) - \mu^2] &= 0 \end{aligned} \tag{6}$$

Solving the quadratic equation, we get

$$\begin{aligned} \sigma_\pm &= \frac{(\omega_1 - i\gamma_1) + (\omega_1 - i\gamma_2)}{2} \pm \frac{\sqrt{[(\omega_1 - i\gamma_1) + (\omega_1 - i\gamma_2)]^2 - 4[(\omega_1 - i\gamma_1)(\omega_2 - i\gamma_2) - \mu^2]}}{2} \\ &= \frac{\omega_1 + \omega_2 - i(\gamma_1 + \gamma_2)}{2} \pm \sqrt{\frac{[(\omega_1 - i\gamma_1) - (\omega_1 - i\gamma_2)]^2}{2} + \mu^2} \end{aligned} \tag{7}$$

Making the substitutions:

$$\begin{aligned} \omega_{\text{ave}} &= \frac{\omega_1 + \omega_2}{2} \qquad & \gamma_{\text{ave}} &= \frac{\gamma_1 + \gamma_2}{2} \\ \omega_{\text{diff}} &= \frac{\omega_1 - \omega_2}{2} \qquad & \gamma_{\text{diff}} &= \frac{\gamma_1 - \gamma_2}{2} \end{aligned} \tag{8}$$

Our eigenvalue expression simplifies to

$$\sigma_\pm = \omega_{\text{ave}} - i\gamma_{\text{ave}} \pm \sqrt{\mu^2 + (\omega_{\text{diff}} - i\gamma_{\text{diff}})^2} \tag{9}$$

And if we solve for the eigenvectors, we get

$$|\sigma_+\rangle = \begin{pmatrix} 1 \\ \frac{\Delta - (\omega_{\text{diff}} - i\gamma_{\text{diff}})}{\mu} \end{pmatrix} \qquad |\sigma_-\rangle = \begin{pmatrix} 1 \\ \frac{-\Delta - (\omega_{\text{diff}} - i\gamma_{\text{diff}})}{\mu} \end{pmatrix} \tag{10}$$

where $\Delta = \sqrt{\mu^2 + (\omega_{\text{diff}} - i\gamma_{\text{diff}})^2}$ is the square root term.
We can see from equations (9) and (10) that if we put the square root term $\Delta = 0$, both the eigenvalues are equal, and so are the eigenvectors, rendering the Hamiltonian undiagonalisable. This phenomenon cannot occur in Hermitian Systems; they can have identical eigenvalues, but their eigenvectors will still be distinct. The point in parameter space where this occurs is called the *Exceptional Point (EP)*.
Let us now find that point in our case.

$$\begin{aligned} \Delta &= 0 \\ \Rightarrow \mu^2 + (\omega_{\text{diff}} - i\gamma_{\text{diff}})^2 &= 0 \\ \Rightarrow (\omega_{\text{diff}} - i\gamma_{\text{diff}})^2 &= -\mu^2 \end{aligned} \tag{11}$$

Since the RHS is real and negative, the LHS must be the square of a purely imaginary number. Thus,

$$\omega_{\text{diff}} = \omega_{\text{EP}} = 0 \qquad \text{and} \qquad \gamma_{\text{diff}} = \gamma_{\text{EP}} = \pm\mu \tag{12}$$

Thus, to reach the EP of the system, the two oscillators must have the same frequency, say ω, and the loss rates must differ by μ in either direction. Let's say the smaller loss rate is γ and the larger one is $\gamma + \mu$. Then, our eigenvalue and eigenvector at EP become:

$$\sigma_{\text{EP}} = \omega - i\Big(\gamma + \frac{\mu}{2}\Big), \qquad |\sigma_{\text{EP}}\rangle = \begin{pmatrix} 1 \\ \pm i \end{pmatrix} \tag{13}$$

The $\pm i$ is based on whether γ_1 is larger or γ_2.
The above example was of a *Second order EP*, where two eigenvalues and eigenvectors coalesced to form a single one. This can also occur in $N \times N$ matrices, where N eigenvalues and eigenvectors coalesce. The EP in that case is called N^{th} *order*.

1.1.1 What makes this useful for sensing?

From our knowledge of sensing at phase transitions, we can connect the dots and infer that sensing near an EP, where the system is vastly different from other nearby points, will give us a larger sensitivity. Let us try and quantify this.
Our first thought should be: "What should we measure, that will give us a good idea of the perturbation's effect?" A good candidate for this is the *eigenvalue splitting*, i.e., the gap between the eigenvalues: $\Delta\sigma = \sigma_+ - \sigma_-$. Frequencies are not hard to measure, and having the difference between two frequencies means that common-mode noise and extra terms are eliminated. Now, let us see how this eigenvalue splitting changes with perturbation.
Let us start with our Hamiltonian

$$\hat{H} = \begin{pmatrix} \omega_1 - i\gamma_1 & \mu \\ \mu & \omega_2 - i\gamma_2 \end{pmatrix}$$

Then, let us add a perturbation of the form $\begin{pmatrix} \epsilon & 0 \\ 0 & 0 \end{pmatrix}$. Our perturbed Hamiltonian is

$$\hat{H}_\epsilon = \begin{pmatrix} \omega_1 + \epsilon - i\gamma_1 & \mu \\ \mu & \omega_2 - i\gamma_2 \end{pmatrix} \tag{14}$$

For now, let us take ϵ to be real and only affecting one oscillator (we will discuss other cases at the end of this part). We can take $\omega_{\text{diff},\epsilon} = \omega_{\text{diff}} + \epsilon/2$.
Now, we know the form of our eigenvalues from (9). Since we only care about $\Delta\sigma$, we evaluate it and find that

$$\Delta\sigma = 2\Delta_\epsilon = 2\sqrt{\mu^2 + (\omega_{\text{diff},\epsilon} - i\gamma_{\text{diff}})^2} = 2\sqrt{\mu^2 + \left(\omega_{\text{diff}} + \frac{\epsilon}{2} - i\gamma_{\text{diff}}\right)^2} \tag{15}$$

From (12), we know that $\omega_{\text{diff}} = 0$ and $\gamma_{\text{diff}} = \pm\mu$, so we can simplify the expression to

$$\begin{aligned}\Delta\sigma &= 2\sqrt{\mu^2 + \left(\frac{\epsilon}{2} \mp i\mu\right)^2} \\ &= 2\sqrt{\cancel{\mu^2} + \frac{\epsilon^2}{4} \mp i\mu\epsilon - \cancel{\mu^2}} \\ &= 2\sqrt{\frac{\epsilon^2}{4} \mp i\mu\epsilon}\end{aligned} \tag{16}$$

Since $\epsilon \ll 1$, we can safely say that ϵ^2 is negligible compared to the ϵ term, giving us

$$\begin{aligned}\Delta\sigma &= 2\sqrt{\mp i\mu\epsilon} = 2(\mu\epsilon\, e^{\mp i\pi/2})^{1/2} \\ &= 2\sqrt{\mu\epsilon}\, e^{\mp i\pi/4} \\ \therefore\ |\Delta\sigma| &= 2\sqrt{\mu\epsilon}\end{aligned} \tag{17}$$

Thus, the magnitude of the splitting is independent of whether γ_1 is less than or greater than γ_2.
So we can see that *the splitting scales with* $\sqrt{\epsilon}$.
For contrast, let us see how the splitting scales away from an EP.

$$\begin{aligned}\Delta\sigma &= 2\sqrt{\mu^2 + \left(\omega_{\text{diff}} + \frac{\epsilon}{2} - i\gamma_{\text{diff}}\right)^2} \\ &= 2\sqrt{\underbrace{\mu^2 + (\omega_{\text{diff}} - i\gamma_{\text{diff}})^2}_{\Delta^2} + (\omega_{\text{diff}} - i\gamma_{\text{diff}})\epsilon + \frac{\epsilon^2}{4}} \\ &= 2\Delta\sqrt{1 + \frac{(\omega_{\text{diff}} - i\gamma_{\text{diff}})\epsilon + \epsilon^2/4}{\Delta^2}}\end{aligned} \tag{18}$$

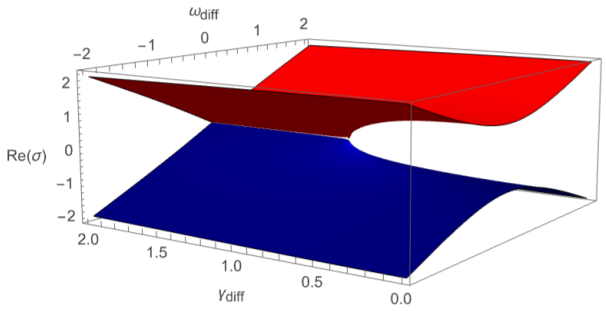
Assuming small ϵ, as we always do, we can ignore the ϵ^2 term, and also use the approximation $\sqrt{1+x} \approx 1 + x/2$, to get

$$\Delta\sigma = 2\Delta\left(1 + \frac{(\omega_{\text{diff}} - i\gamma_{\text{diff}})\epsilon}{2\Delta^2}\right) = 2\Delta + \frac{(\omega_{\text{diff}} - i\gamma_{\text{diff}})\epsilon}{\Delta} \tag{19}$$

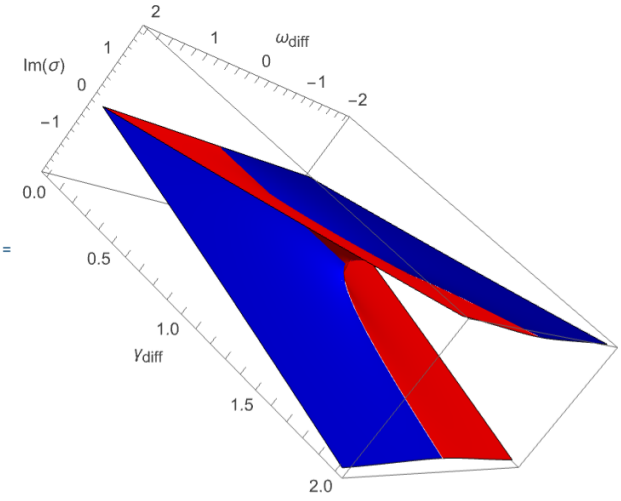
Thus, the difference of this splitting from the unperturbed case ($\Delta\sigma = 2\Delta$) is *proportional to* ϵ.
We can therefore see the difference in splitting between the EP and non-EP case. The EP case has its splitting scaling with $\sqrt{\epsilon}$, which is better in the small ϵ case, which is where we operate.
Using a similar process, we could show that for an Nth order EP, the splitting scales as $\epsilon^{1/N}$.
Therefore, we have proved that sensing at exceptional points provides a significant enhancement in precision, when compared to sensing outside exceptional points.

1.1.2 Visualisation

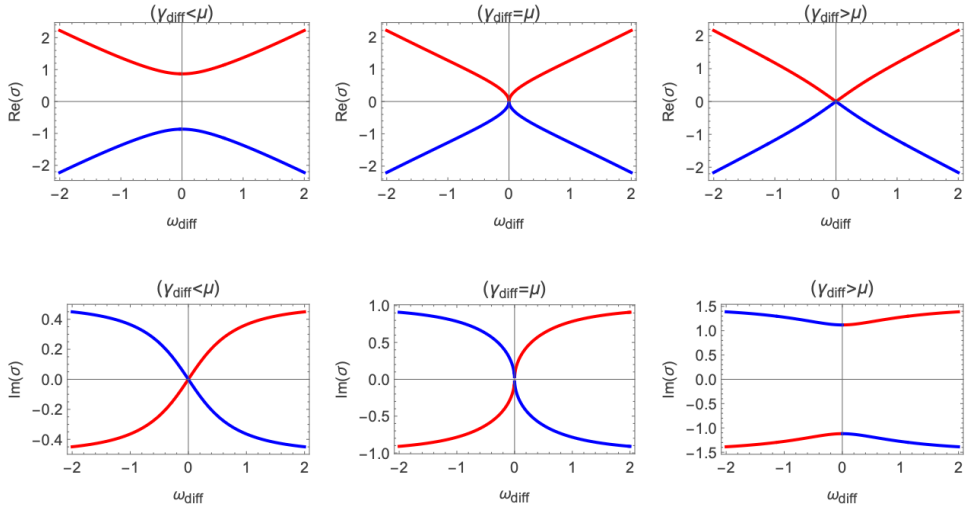
Let us now make a plot that can show us how the splitting behaves at and around our exceptional point. We will need to fix some parameters as constant in order to have a visualisable plot. Thus, let us fix $\omega_{\text{ave}} = 0$ and $\gamma_{\text{ave}} = 0$ and make a 3D plot of Equation (9) with ω_{diff} and γ_{diff} as the axes.
Refer Figure 1 for these visualisations.



(a) Real part of the two eigenvalues



(b) Imaginary part of the two eigenvalues



(c) Eigenvalues vs ω_{diff} for 3 different γ_{diff} cases

Figure 1: **(a) and (b)***: 3D Plots of the eigenvalues. Note how they make contact at* $\omega_{diff} = 0$ *and* $\gamma_{diff} = \mu = 1$*. We can intuitively tell that this is the exceptional point.* **(c)***: Real and Imaginary parts of the Eigenvalues for 3 cases: before, at and after the EP.*
For all parts, note how they make contact vertically at $\omega_{diff} = 0$ *and* $\gamma_{diff} = \mu = 1$*. We can intuitively tell that this is the exceptional point.*

1.2 Parity-Time (PT) Symmetry

A system is PT Symmetric if it is invariant under the action of the parity operator $\hat{\mathcal{P}}$ **and** the time operator $\hat{\mathcal{T}}$, which are defined as follows:

- Parity $\hat{\mathcal{P}}$: reversal of spatial coordinates ($x \to -x$). In the 2×2 case, it can be represented as the Pauli-X matrix σ_x.
- Time $\hat{\mathcal{T}}$: reversal of time ($t \to -t$). In Quantum Mechanics, we prefer to do $i \to -i$ instead (Complex Conjugation Operator). This is because it is able to perform time-reversal in equations where time is not *explicitly* mentioned. For example, the definition of the momentum operator $\hat{p} = -i\hbar\frac{\partial}{\partial x}$.

A direct mathematical way to check if the system is PT-symmetric, is to see if its Hamiltonian commutes with $\hat{\mathcal{P}}\hat{\mathcal{T}}$. If it is PT-symmetric,

$$[\hat{H}, \hat{\mathcal{P}}\hat{\mathcal{T}}] = 0 \qquad \text{or} \qquad \hat{\mathcal{P}}\hat{\mathcal{T}}\hat{H}(\hat{\mathcal{P}}\hat{\mathcal{T}})^{-1} = \hat{H} \tag{20}$$

Let us check PT-symmetry in our Hamiltonian in (5).

$$\begin{aligned}
\hat{\mathcal{P}}\hat{\mathcal{T}}\hat{H}(\hat{\mathcal{P}}\hat{\mathcal{T}})^{-1} &= \hat{\mathcal{P}}\hat{\mathcal{T}}\hat{H}\hat{\mathcal{T}}^{-1}\hat{\mathcal{P}}^{-1} \\
&= \hat{\mathcal{P}}\hat{\mathcal{T}}\begin{pmatrix}\omega_1 - i\gamma_1 & \mu \\ \mu & \omega_2 - i\gamma_2\end{pmatrix}\hat{\mathcal{T}}^{-1}\hat{\mathcal{P}}^{-1} \\
&= \sigma_x\begin{pmatrix}\omega_1 + i\gamma_1 & \mu \\ \mu & \omega_2 + i\gamma_2\end{pmatrix}\sigma_x = \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}\begin{pmatrix}\omega_1 + i\gamma_1 & \mu \\ \mu & \omega_2 + i\gamma_2\end{pmatrix}\begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix} \\
&= \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}\begin{pmatrix}\mu & \omega_1 + i\gamma_1 \\ \omega_2 + i\gamma_2 & \mu\end{pmatrix} \\
\therefore \hat{\mathcal{P}}\hat{\mathcal{T}}\hat{H}(\hat{\mathcal{P}}\hat{\mathcal{T}})^{-1} &= \begin{pmatrix}\omega_2 + i\gamma_2 & \mu \\ \mu & \omega_1 + i\gamma_1\end{pmatrix}
\end{aligned} \tag{21}$$

Let us compare the two:

$$\hat{H} = \begin{pmatrix}\omega_1 - i\gamma_1 & \mu \\ \mu & \omega_2 - i\gamma_2\end{pmatrix}, \qquad \hat{\mathcal{P}}\hat{\mathcal{T}}\hat{H}(\hat{\mathcal{P}}\hat{\mathcal{T}})^{-1} = \begin{pmatrix}\omega_2 + i\gamma_2 & \mu \\ \mu & \omega_1 + i\gamma_1\end{pmatrix}$$

Therefore, for the system to be PT-symmetric, we need

$$\omega_1 = \omega_2 \qquad \text{and} \qquad \gamma_1 = -\gamma_2 \tag{22}$$

1.2.1 Visualisation

Combining the conditions in (12) and (22), the conditions we require to get an EP in a PT-symmetric system are

$$\omega_1 = \omega_2 \qquad \gamma_1 = -\gamma_2 \quad \text{and} \quad \gamma_1 - \gamma_2 = \pm\mu \tag{23}$$

In other words,

$$\omega_{\text{ave}} = \omega \qquad \omega_{\text{diff}} = 0 \qquad \gamma_{\text{ave}} = 0 \quad \text{and} \quad \gamma_{\text{diff}} = \pm\mu \tag{24}$$

ω_{diff} being 0 has removed an extra dimension from our plot, so our earlier 3D plots in Figures 1a and 1b become 2D plots in Figure 2. Thus, we see that before the EP, the eigenvalues are real and distinct, known as "Exact PT". After the EP, the eigenvalues are imaginary and distinct, known as "Broken PT". Only at the EP are the eigenvalues equal.

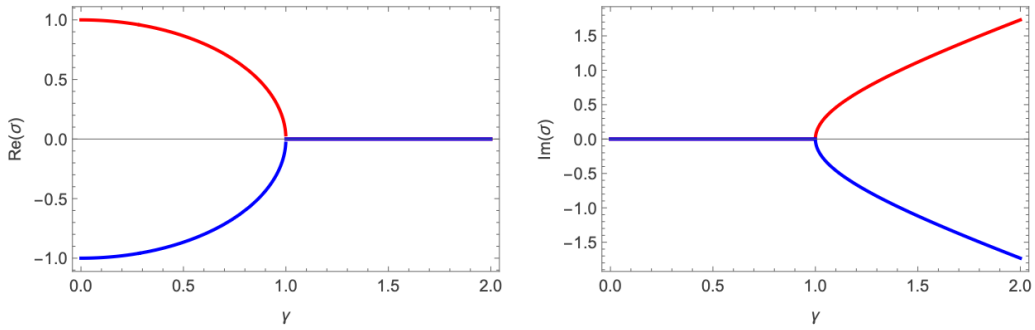


Figure 2: Real (left) and Imaginary (right) parts of the eigenvalues of a PT-symmetric system, changing with $\gamma_1 = -\gamma_2 = \gamma$.

1.2.2 Why do we want PT-Symmetry?

We know that we can combine the conditions in (12) and (22) to get a PT-Symmetric System with an Exceptional Point. However, what advantages does PT-Symmetry bring from a sensing perspective?

- **It makes the EP crossing a Phase Transition:** Without PT-symmetry, the eigenvalues on either side of the EP are complex. However, in a PT-symmetric system, the eigenvalues are purely real below the EP, and purely imaginary above it. This means upon crossing the EP, the system transitions from oscillatory ($e^{i\sigma t}$) to growing/decaying ($e^{\pm\sigma t}$). Thus, the transition is more drastic, and easier to observe.
- **It makes finding the EP easier:** Since $\gamma_1 = -\gamma_2 = \gamma$, i.e., equal gain and loss, We only need to control γ to effectively control both. This makes it easier to visualise finding the EP, as made evident by the removal of a dimension in our plots in Figure 2.

For these advantages, we will be sensing at the EP of a PT-symmetric system.

2 Experimental Setup

Now, let us begin our discussion of the sensing experiment done in [1].
From our discussion so far, we have understood that to get a PT-symmetric system, we need two oscillators that have the same natural frequency, with equal and opposite gain/loss. To realise this, we will make the circuit shown in Figure 3, with two LC oscillators ($L_1 = L_2 = L, C_1 = C_2 = C$). There are also resistors in parallel with them, for gain/loss. In order to realise gain, the resistance in the gain block will be negative.

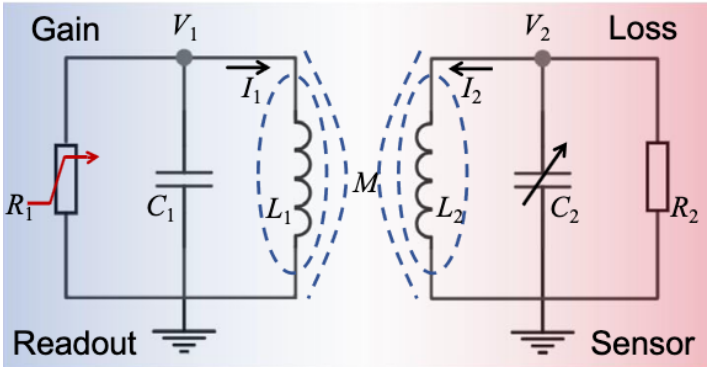


Figure 3: PT-symmetric system realised with LC oscillators. The resistance in the gain block is negative. M is the mutual inductance between the inductors.

Let us define some variables before we find this system's Hamiltonian. Defined the "normalised time" as $\tau = \omega_0 t$, where $\omega_0 = 1/\sqrt{LC}$. Define the "normalised frequency" as $\lambda = \omega/\omega_0$. Then the two node voltages can be written as $\mathbf{v_1} = V_1 e^{i\omega t} = V_1 e^{i\lambda\tau}$ and $\mathbf{v_2} = V_2 e^{i\lambda\tau}$. Similarly, the two branch currents can be written as $\mathbf{i_1} = I_1 e^{i\lambda\tau}$ and $\mathbf{i_2} = I_2 e^{i\lambda\tau}$. The coupling constant is defined as $k = M/L$. The gain and loss coefficients are defined as follows:

$$g = \frac{1}{R_1}\sqrt{\frac{L}{C}} \qquad \gamma = \frac{1}{R_2}\sqrt{\frac{L}{C}}$$

Note that even though the resistance in the gain block is negative, we will take R_1 and g to be positive quantities in our calculations, and adjust the signs accordingly.
Let us write KCL equations for the two oscillators:

$$\begin{aligned} \mathbf{i_1} + C\dot{\mathbf{v}}_\mathbf{1} - \mathbf{v_1}/R_1 = 0 \\ \mathbf{i_2} + C\dot{\mathbf{v}}_\mathbf{2} + \mathbf{v_2}/R_2 = 0 \end{aligned} \tag{25}$$

Here, $\dot{\mathbf{v}}_\mathbf{n} = d\mathbf{v_n}/dt$. Since $\mathbf{v_n} = V_n e^{i\omega t}$, we can say that $\dot{\mathbf{v}}_\mathbf{n} = i\omega\mathbf{v_n}$, and our equations become

$$\begin{aligned} \mathbf{i_1} + iC\omega\mathbf{v_1} - \mathbf{v_1}/R_1 = 0 \\ \mathbf{i_2} + iC\omega\mathbf{v_2} + \mathbf{v_2}/R_2 = 0 \end{aligned} \tag{26}$$

Now, using inductance relations $\mathbf{v_n} = L\frac{d\mathbf{i_n}}{dt} = i\omega L\mathbf{i_n}$, we get the equations

$$\begin{aligned} \mathbf{v_1} = i\omega L\mathbf{i_1} + i\omega kL\mathbf{i_2} \\ \mathbf{v_2} = i\omega kL\mathbf{i_1} + i\omega L\mathbf{i_2} \end{aligned} \tag{27}$$

Recall that $k = M/L \Rightarrow kL = M$.
We can now substitute (26) into (27) to get the following equations. We will just deal with the first equation of the pair for now.

$$\begin{aligned} \mathbf{v_1} &= i\omega L\mathbf{i_1} + i\omega kL\mathbf{i_2} \\ &= i\omega L(-iC\omega\mathbf{v_1} + \mathbf{v_1}/R_1) + i\omega kL(-iC\omega\mathbf{v_2} - \mathbf{v_2}/R_2) \\ &= \omega^2 LC\mathbf{v_1} + i\omega(L/R_1)\mathbf{v_1} + \omega^2 kLC\mathbf{v_2} - i\omega k(L/R_2)\mathbf{v_2} \\ &= \lambda^2 \cancel{\omega_0^2}\frac{1}{\cancel{\omega_0^2}}\mathbf{v_1} + i\frac{\lambda}{\sqrt{LC}}\frac{L}{R_1}\mathbf{v_1} + k\lambda^2\cancel{\omega_0^2}\frac{1}{\cancel{\omega_0^2}}\mathbf{v_2} - ik\frac{\lambda}{\sqrt{LC}}\frac{L}{R_2}\mathbf{v_2} \\ &= \lambda^2\mathbf{v_1} + i\lambda\frac{1}{R_1}\sqrt{\frac{L}{C}}\mathbf{v_1} + k\lambda^2\mathbf{v_2} - ik\lambda\frac{1}{R_2}\sqrt{\frac{L}{C}}\mathbf{v_2} \\ \therefore \mathbf{v_1} &= \lambda^2\mathbf{v_1} + ig\lambda\mathbf{v_1} + k\lambda^2\mathbf{v_2} - ik\gamma\lambda\mathbf{v_2} \end{aligned} \tag{28}$$

We can do the same process on the second equation to finally obtain the pair

$$\begin{aligned}\mathbf{v_1} &= \lambda^2\mathbf{v_1} + ig\lambda\mathbf{v_1} + k\lambda^2\mathbf{v_2} - ik\gamma\lambda\mathbf{v_2} \\ \mathbf{v_2} &= k\lambda^2\mathbf{v_1} + ikg\lambda\mathbf{v_1} + \lambda^2\mathbf{v_2} - i\gamma\lambda\mathbf{v_2}\end{aligned} \tag{29}$$

Further simplifications are required, so let us just deal with the first equation for now

$$\begin{aligned}\mathbf{v_1} &= \lambda^2\mathbf{v_1} + ig\lambda\mathbf{v_1} + k\lambda^2\mathbf{v_2} - ik\gamma\lambda\mathbf{v_2} \\ (ig\lambda + \lambda^2 - 1)\mathbf{v_1} &+ (-ik\gamma\lambda + k\lambda^2)\mathbf{v_2} = 0\end{aligned} \tag{30}$$

The system will be near resonance all the time, so $\lambda \approx 1$. Thus, we can Taylor expand λ around 1. Upto first order, that expansion is $\lambda^2 \approx 1 + 2(\lambda - 1) = 2\lambda - 1$. Substituting this in, we get

$$\begin{aligned}(ig\lambda + 2\lambda - 1 - 1)\mathbf{v_1} + (-ik\gamma\lambda + k(2\lambda - 1))\mathbf{v_2} &= 0 \\ \Rightarrow (ig\lambda + 2\lambda - 2)\mathbf{v_1} + (-ik\gamma\lambda + 2k\lambda - k)\mathbf{v_2} &= 0\end{aligned} \tag{31}$$

Now, R_1 and R_2 will be large values, so as not to damp the oscillator too much. As a result, $g, \gamma \ll 1$. In addition, $k = M/L$ will also be $\ll 1$, since mutual inductance in typical cases is much less than self-inductance. Now we have established that g, γ and k are all $\ll 1$. Thus, the $-ik\gamma\lambda$ term in the above equation can be regarded as second order and thus ignored. This leaves us with the equation

$$\begin{aligned}(ig\lambda + 2\lambda - 2)\mathbf{v_1} + (2k\lambda - k)\mathbf{v_2} &= 0 \\ \Rightarrow 2\mathbf{v_1} + k\mathbf{v_2} &= \lambda[(2 + ig)\mathbf{v_1} + 2k\mathbf{v_2}] \\ \Rightarrow \mathbf{v_1} + \frac{k}{2}\mathbf{v_2} &= \lambda\left[\left(1 + i\frac{g}{2}\right)\mathbf{v_1} + k\mathbf{v_2}\right]\end{aligned} \tag{32}$$

Now since k is small and $\lambda - 1 = \Delta\lambda$ is also small, the $k \cdot \Delta\lambda$ terms can be neglected. Thus, we can say that $k\lambda \approx k \cdot 1$. Similarly, $ig\lambda/2 \approx (ig \cdot 1)/2$. This gives us

$$\begin{aligned}\mathbf{v_1} + \frac{k}{2}\mathbf{v_2} &= \lambda\mathbf{v_1} + i\frac{g}{2}\mathbf{v_1} + k\mathbf{v_2} \\ \therefore \left(1 - i\frac{g}{2}\right)\mathbf{v_1} - \frac{k}{2}\mathbf{v_2} &= \lambda\mathbf{v_1}\end{aligned} \tag{33}$$

Using the same method, we can simplify the second equation in (29) to get

$$-\frac{k}{2}\mathbf{v_1} + \left(1 + i\frac{\gamma}{2}\right)\mathbf{v_2} = \lambda\mathbf{v_2} \tag{34}$$

Therefore, combining the previous two equations, we get the following matrix equation

$$\begin{pmatrix} 1 - i\frac{g}{2} & -\frac{k}{2} \\ -\frac{k}{2} & 1 + i\frac{\gamma}{2} \end{pmatrix} \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \end{pmatrix} = \lambda \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \end{pmatrix} \tag{35}$$

Which is in the form of an eigenvalue equation (We can also replace the $\mathbf{v_n}$ in this equation with V_n if we like, since the common $e^{i\lambda\tau}$ factor will cancel).
Using the fact that $d\mathbf{v_n}/d\tau = i\lambda\mathbf{v_n}$, we finally obtain

$$\boxed{-i\frac{d}{d\tau}\begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \end{pmatrix} = \begin{pmatrix} 1 - i\frac{g}{2} & -\frac{k}{2} \\ -\frac{k}{2} & 1 + i\frac{\gamma}{2} \end{pmatrix} \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \end{pmatrix}} \tag{36}$$

which is our system's equivalent to (2).
We can use the eigenvalue equation (35) to obtain the characteristic equation

$$\begin{aligned}\det(H - \lambda I) &= 0 \\ \Rightarrow \det\begin{pmatrix} 1 - \lambda - i\frac{g}{2} & -\frac{k}{2} \\ -\frac{k}{2} & 1 - \lambda + i\frac{\gamma}{2} \end{pmatrix} &= 0 \\ \Rightarrow \left(1 - \lambda - i\frac{g}{2}\right)\left(1 - \lambda + i\frac{\gamma}{2}\right) - \frac{k^2}{4} &= 0 \\ \Rightarrow (1 - \lambda)^2 + (1 - \lambda)\frac{i}{2}(\gamma - g) + \frac{g\gamma}{4} - \frac{k^2}{4} &= 0\end{aligned} \tag{37}$$

For the PT-symmetric case, we know that gain and loss rates are equal, meaning $g = \gamma$. This simplifies our characteristic equation to

$$(1-\lambda)^2 + \frac{\gamma^2}{4} - \frac{k^2}{4} = 0 \tag{38}$$

Which we can easily solve to get the eigenvalues

$$\lambda_\pm = 1 \pm \frac{1}{2}\sqrt{k^2 - \gamma^2} \tag{39}$$

Therefore the system is in Exact Phase with 2 real eigenvalues when $\gamma < k$, and in Broken Phase with 2 complex eigenvalues (real part stays constant, imaginary part changes with γ) when $\gamma > k$. At $\gamma = k$, the eigenvalues and eigenvectors coalesce, making it an exceptional point. $\lambda_{\text{EP}} = 1$.
Thus, we have assembled the system we will use for sensing. Now to perturb it and see how the system responds.

3 Perturbing the System

Now let us perturb the system, then allow it to settle to a new steady state, and see how much the eigenfrequencies have shifted compared to the unperturbed case.
We will perturb the capacitor in the loss circuit C_2 (hence the arrow across it in Figure 3). Let the perturbation be ΔC, meaning relative perturbation is $\delta = \Delta C/C_2 = \Delta C/C$. We will enforce that $0 < \delta \ll 1$. After a few cycles, the system has settled into a new steady state of oscillation. Its equation is now

$$\begin{pmatrix} 1-\lambda-i\frac{g'}{2} & -\frac{k}{2} \\ -\frac{k}{2} & 1-\lambda-\frac{\delta}{2}+i\frac{\gamma}{2} \end{pmatrix} \begin{pmatrix} V_1' \\ V_2' \end{pmatrix} = 0 \tag{40}$$

Where V_1' and V_2' are the new node voltages of the new steady state. Since the gain depends on a non-linear resistor, this will also cause the gain to becomes g'.
How does this $-\delta/2$ term come? It comes from the change in the natural frequency of the second resonator.

$$\begin{aligned} \omega_2 &= \frac{1}{\sqrt{LC(1+\delta)}} = (1+\delta)^{-1/2}\omega_0 \\ \omega_2 &\approx \omega_0\left(1-\frac{\delta}{2}\right) \\ \therefore \frac{\omega_2}{\omega_0} &= 1-\frac{\delta}{2} \end{aligned} \tag{41}$$

Recall that in the unnormalised equation (2), the diagonal terms were the resonant frequncies ω_1 and ω_2. They became 1 upon normalisation because both were equal to ω_0. Now the 1 in the bottom right term will be changed to $1-\delta/2$, thus giving us our new matrix equation.
Why is the effect of the perturbation on γ not shown in the new equation? that is because the extra term that comes from the change in γ will have $\gamma\delta$ in it. Since both of these are small values, they can be ignored.
To clarify, the λ here will also be different from the unperturbed case, since it is the eigenvalue for a new matrix. Let us now solve for that.

$$\begin{aligned} \det\begin{pmatrix} 1-\lambda-i\frac{g'}{2} & -\frac{k}{2} \\ -\frac{k}{2} & 1-\lambda-\frac{\delta}{2}+i\frac{\gamma}{2} \end{pmatrix} &= 0 \\ \Rightarrow \left(1-\lambda-i\frac{g'}{2}\right)\left(1-\lambda-\frac{\delta}{2}+i\frac{\gamma}{2}\right) - \frac{k^2}{4} &= 0 \\ \Rightarrow (1-\lambda)^2 + \frac{\lambda\delta}{2} - \frac{\delta}{2} - \frac{k^2}{4} + \frac{\gamma g'}{4} + i\left(-\frac{g'}{2} + \frac{\delta g'}{4} + \frac{\gamma}{2} + \frac{\lambda g'}{2} - \frac{\lambda\gamma}{2}\right) &= 0 \end{aligned} \tag{42}$$

We have two unknown: g' and λ, and two equations, since the real and imaginary parts must be 0 individually. Let us start with the imaginary part.

$$\begin{aligned} -\frac{g'}{2} + \frac{\delta g'}{4} + \frac{\gamma}{2} + \frac{\lambda g'}{2} - \frac{\lambda\gamma}{2} &= 0 \\ \Rightarrow \left(\lambda - 1 + \frac{\delta}{2}\right)g' + (1-\lambda)\gamma &= 0 \\ \therefore g' &= \frac{2\gamma(\lambda-1)}{2(\lambda-1)+\delta} \end{aligned} \tag{43}$$

Now we can equate the real part of (42) to zero and substitute this expression in to get

$$(1-\lambda)^2 + \frac{\lambda\delta}{2} - \frac{\delta}{2} - \frac{k^2}{4} + \frac{\gamma}{4}\frac{2\gamma(\lambda-1)}{2(\lambda-1)+\delta} = 0 \tag{44}$$

This is a cubic equation in λ. Let us expand it out fully.

$$\begin{aligned}(1-\lambda)^2 + \frac{(\lambda-1)\delta}{2} - \frac{k^2}{4} + \frac{1}{2}\frac{\gamma^2(\lambda-1)}{2(\lambda-1)+\delta} = 0 \\ \Rightarrow 4(\lambda-1)^2[2(\lambda-1)+\delta] + 2(\lambda-1)\delta\left[2(\lambda-1)+\delta\right] - k^2[2(\lambda-1)+\delta] + 2\gamma^2(\lambda-1) = 0\end{aligned}$$

Let $(\lambda - 1) = t$.

$$\begin{aligned}8t^3 + 4\delta t^2 + 4\delta t^2 + 2\delta^2 t - 2k^2 t - k^2\delta + 2\gamma^2 t = 0 \\ \Rightarrow 8t^3 + 8\delta t^2 + (2\delta^2 - 2k^2 + 2\gamma^2)t - k^2\delta = 0\end{aligned} \tag{45}$$

We can use *Cardano's method* to solve this cubic. Its first step is to reduce it to the "depressed cubic form", which is done by making the substitution $t = x - b/3a$. In our case, the substitution is $t = x - \delta/3$. This gives us

$$\begin{aligned}8\left(x-\frac{\delta}{3}\right)^3 + 8\delta\left(x-\frac{\delta}{3}\right)^2 + 2(\delta^2 - k^2 + \gamma^2)\left(x-\frac{\delta}{3}\right) - k^2\delta = 0 \\ \Rightarrow \left(x^3 - \cancel{\delta x^2} + \frac{\delta^2}{3}x - \frac{\delta^3}{27}\right) + \delta\left(\cancel{x^2} - \frac{2\delta x}{3} + \frac{\delta^2}{9}\right) + \left(\frac{\delta^2 - k^2 + \gamma^2}{4}\right)x - \frac{\delta^3 - \delta k^2 + \delta\gamma^2}{12} - \frac{k^2\delta}{8} = 0 \\ \Rightarrow x^3 + \left(\frac{\delta^2}{3} - \frac{2\delta^2}{3} + \frac{\delta^2 - k^2 + \gamma^2}{4}\right)x - \frac{\delta^3}{27} + \frac{\delta^3}{9} - \frac{\delta^3 - \delta k^2 + \delta\gamma^2}{12} - \frac{k^2\delta}{8} = 0 \\ \Rightarrow x^3 + \left(\frac{-\delta^2 - 3k^2 + 3\gamma^2}{12}\right)x - \frac{-8\delta^3 + 24\delta^3 - 18\delta^3 + 18\delta k^2 - 18\delta\gamma^2 - 27\delta k^2}{216} = 0 \\ \Rightarrow x^3 + \left(\frac{-\delta^2 - 3k^2 + 3\gamma^2}{12}\right)x - \frac{-2\delta^3 - 9\delta k^2 - 18\delta\gamma^2}{216} = 0\end{aligned} \tag{46}$$

Fitting this with the $x^3 + px + q = 0$ form, we get

$$p = \frac{-\delta^2 - 3k^2 + 3\gamma^2}{12} \qquad q = \frac{-2\delta^3 - 9\delta k^2 - 18\delta\gamma^2}{216}$$

We can use Cardano's discriminant $\Delta = -4p^3 - 27q^2$ to see that this equation has one real and two complex roots. Now, the supplementary material in [1] defines A, B and C as follows:

$$A = -16\delta^2 - 48k^2 + 48\gamma^2 \qquad B = (C + \sqrt{4A^3 + C^2})^{1/3} \qquad C = 128\delta^3 + 576\delta k^2 + 1152\delta\gamma^2$$

Comparing A and C with p and q, we get that

$$p = \frac{A}{192} \qquad q = -\frac{C}{13824}$$

thus giving us the equation

$$x^3 + \left(\frac{A}{192}\right)x - \frac{C}{13824} = 0 \tag{47}$$

Now, from Cardano's formula, the real root is given by

$$x = \underbrace{\left(-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}\right)^{1/3}}_{u} + \underbrace{\left(-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}\right)^{1/3}}_{v} \tag{48}$$

Let us evaluate each term separately.

$$\begin{aligned}u^3 &= \frac{C}{2\times 13824} + \sqrt{\frac{C^2}{4\times 13824^2} + \frac{A^3}{27\times 192^3}} \\ &= \frac{C}{27648} + \sqrt{\frac{C^2 + 4A^3}{4\times 13824^2}} = \frac{C}{27648} + \frac{\sqrt{C^2+4A^3}}{27648} \\ \Rightarrow u &= \frac{(C + \sqrt{C^2+4A^3})^{1/3}}{(24^3\times 2)^{1/3}} = \frac{B}{24\times 2^{1/3}}\end{aligned} \tag{49}$$

Now, if we substitute x with $u+v$ in the cubic, we are making it a cubic in two variables instead of one. In order to preserve the degrees of freedom, we must enforce an extra constraint on u and v. Upon expanding the substituted cubic, we would have

$$\begin{aligned}(u^3+v^3+3uv(u+v))+p(u+v)+q=0\\ \Rightarrow u^3+v^3+q+(3uv+p)(u+v)=0\end{aligned}$$

In order to make the $(u+v)$ term vanish, we can enforce the constraint $(3uv+p)=0 \Rightarrow uv=-p/3$. We can use this to find v.

$$\begin{aligned}u\cdot v&=-\frac{p}{3}\\ \Rightarrow v&=-\frac{p}{3u}=\frac{-A/192}{3}\frac{24\times 2^{1/3}}{B}\\ v&=-\frac{A}{12\cdot 2^{2/3}B}\end{aligned}\tag{50}$$

Therefore, the real root is

$$\begin{aligned}x&=\frac{B}{24\times 2^{1/3}}-\frac{A}{12\cdot 2^{2/3}B}\\ \Rightarrow t&=-\frac{\delta}{3}+\frac{B}{24\times 2^{1/3}}-\frac{A}{12\cdot 2^{2/3}B}\\ \Rightarrow \lambda&=\frac{3-\delta}{3}+\frac{B}{24\times 2^{1/3}}-\frac{A}{12\cdot 2^{2/3}B}\end{aligned}\tag{51}$$

Now, to get the root in terms of our earlier parameters, we will need to make some approximations in the $\delta\to 0$ limit. Refer the supplementary material of [1] for this derivation.

Final answer:
The real root is

$$\lambda\approx 1+\frac{k^{2/3}}{2}\delta^{1/3}\tag{52}$$

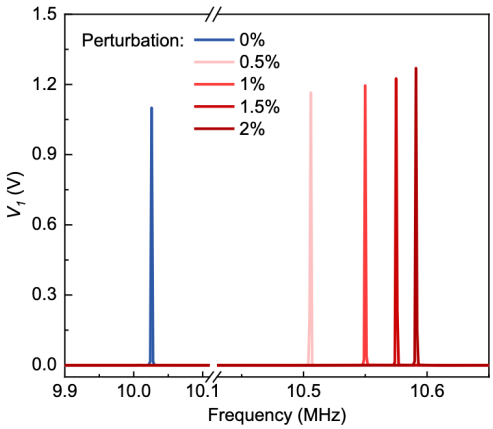
We can do a quick sanity check by putting $\delta=0$. This gives us $\lambda=1$, which is the value of λ_{EP} in the unperturbed case. This is an unexpected result. We had shown in Section 1.1.1 that our measurement scales as the square root of the perturbation, but here we are obtaining the cube root.
The reason behind this is the non-linearity of our system. If we take a look at the perturbed Hamiltonian in Section 1.1.1, the perturbation ϵ affects only the first oscillator and nothing else. In our system, since the gain is non-linear, the perturbation in the second oscillator causes an adjustment in gain of the first oscillator in order to reach a new steady state.
Say our system was linear, so $g'=g$, which is known. Then the characteristic equation (42) has highest order 2, meaning it will depend on $\delta^{1/2}$. It is thanks to the non-linearity of the system that we obtain (43), the expression for g', having λ in its denominator, thus making the equation a cubic one.

4 Dealing with Noise

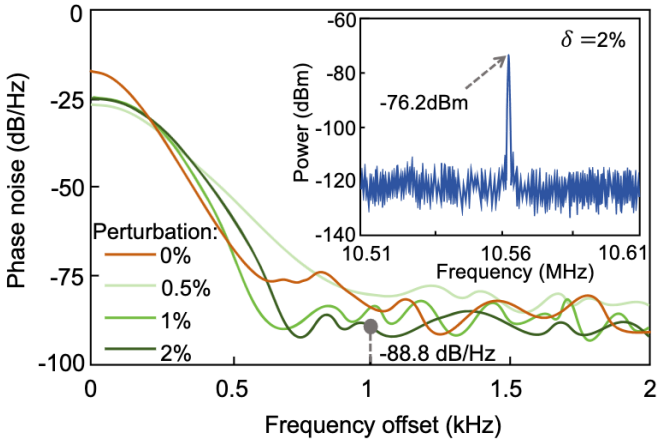
EP sensors have recently been critiqued for not offering any fundamental enhancement over other similar sensors [3]. Let us see if and how [1] overcomes this.
As we just established, in the linear case, our characteristic equation is a quadratic in λ, so it yields two possible eigenvalues. When a perturbation causes the system to go beyond the EP, these eigenvalues become complex conjugates. Since eigenvalues appear as $e^{i\lambda\tau}$, the imaginary part of the eigenvalues manifests as gain/loss ($e^{i\lambda_r\tau}\cdot e^{-\lambda_i\tau}$). (Note that the eigenfrequency just means the real part of the eigenvalue.) When making the spectrum of this, the peak at λ_r will have Full Width at Half Maxima (FWHM) equal to $2|\lambda_i|$. Thus, we see that the imaginary part causes "line broadening". This is a form of noise as it makes our eigenfrequency measurement less precise.
However, the non-linearity of our system helps with this problem. Since the non-linearity causes our characteristic equation to be cubic, we now have an *extra real root* along with our two complex conjugate roots. If we find the new gain g' corresponding to each of these eigenfrequencies, we will find that the real eigenvalue is the one that requires the least gain. Thus, it will dominate over the other two. This makes the spectrum peak much sharper than the linear case, significantly reducing the noise introduced by the perturbation.
We can quantify this ratio of sensitivity to line broadening using the Figure of Merit (FOM), which in our case is defined as

$$\text{FOM}=20\log_{10}\frac{\partial(\Delta f)/\partial\delta}{\text{FWHM}}\tag{53}$$

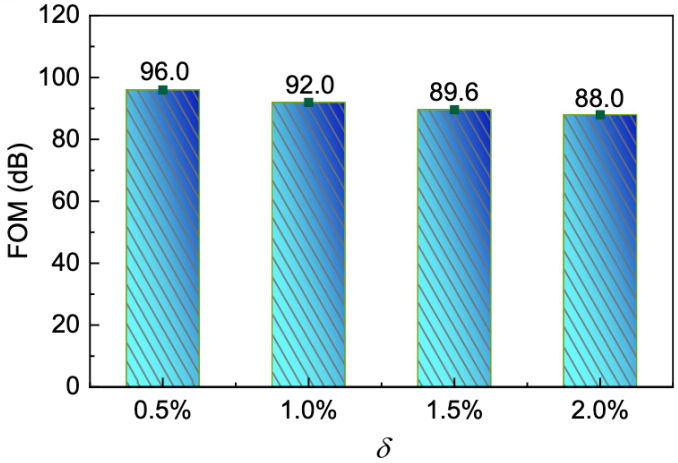
We can see the frequency peak shifting based on perturbation in Figure 4a, and how the FOM changes with perturbation in Figure 4c.
Figure 4b shows the phase noise plotted against the "frequency offset" from the resonant frequency. In its inset, we have the signal power plotted against the frequency. [1] states that the FWHM of the peak in the plot is 0.2kHz, which when compared to the frequency shift of 0.54MHz, is very small.
Amplitude noise is also successfully suppressed thanks to the non-linear gain. If amplitude increases, the gain will decrease and vice versa, thus forcing the amplitude to not fluctuate.



(a) Showing how the frequency shift changes with perturbation from 0% to 2%. It is obtained by performing an FFT on the final node voltage after the system has settled [1].



(b) The phase noise plotted against the frequency offset from the resonant frequency, for different perturbations from 0% to 2%. The inset shows the power of the output signal plotted against the absolute frequency, for 2% perturbation [1].



(c) The measured Figure of Merit for 0% to 2% perturbation [1].

Figure 4

References

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